Rajiv Gandhi University of Knowledge Technologies - RK Valley -AP

Academic Year 2024 – 25: Semester II - Mid Semester Test-1

Computer Science & Engineering - 23CS1201- Object Oriented Programming Through Java

R22 Batch & Regular Mode of Examinations

Year: ENGG-1 | Date of Examination: 31st Jan, 2025

Time: 60 Minutes R220714 | Max Marks: 15

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3X5 Marks= 15

The Question Paper consists of 3 units. Answer one full question from each unit.
Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions

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1. What are the Key features of Java? Explain about them

**OR**

2. Explain about the fundamentals of Class and Object with sample code

3. Explain about the different types of Variables and its scope with sample code.

**OR**

4. What are the different Access specifiers? Explain about all Access specifiers with sample code

5. Explain about the following concepts?
   a. Constructors (3 Marks)
   b. Wrapper classes ( 2 Marks)

**OR**

6. Explain about the String class and explain the String methods with sample code

Rajiv Gandhi University of Knowledge Technologies - RK Valley - AP

Academic Year 2024– 25: Semester 1 - Mid Semester Test-1

Computer Science & Engineering - Data strcures

R22Batch & Regular Mode of Examinations

Year: ENGG-1 | Date of Examination: 30-01-2025

Time: 60 Minutes R220711 | Max Marks: 15

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3X5 Marks= 15

The Question Paper Consists of 6 questions in 3 sets. Students need to answer 1 question from each set. Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions

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1. Explain Singly Linked list deletion operations of all 3 cases.

**(OR)**

2. Explain any 5 differences between arrays and linkedlist.

3. Explain Doubly Linked list insertion operations of all 3 cases.

**(OR)**

4. Explain Circular Singly Linked list insertion operations of all 3 cases.

5. Explain stack operations using arrays in detail.

**(OR)**

6. Explain Circular doubly Linked list insertion operations of all 3 cases

# Rajiv Gandhi University of Knowledge Technologies - RK Valley- AP

Academic Year 2024 – 25: Semester 2 - Mid Semester Test-1

Computer Science & Engineering - 23PY1201 – Engineering Physics

R22 Batch & Regular Mode of Examinations

Year: ENGG-1 | Date of Examination: 1st Feb, 2025.

Time: 60 Minutes R220711 | Max Marks: 15

3X5 Marks= 15

The Question Paper consists of 6 questions in 3 sets. Students need to answer 1 question from each set. Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions

Note: Usage of Scientific calculator may be permitted

1. Define the Simple Harmonic Oscillation (SHM). Derive the solution of SHM and discuss the phase relationship between displacement, velocity and acceleration with neat diagram. 5M

(OR)

2. (a) What are damped oscillations? Derive the solution of damped oscillation and mention the types of damped Oscillations with examples. 4M

(b) A frequency of a tuning fork is 500 Hz, its quality factor is 30000. Find the time after which its energy becomes $\frac{1}{10}$ th of its initial value. 1M

3. (a) What are the forced oscillations and derive the solution for *displacement of* weekly damped forced oscillator. 4M

(b) A body of mass 0.2 kg is suspended from a spring of force constant 80 $Nm^{-1}$. A damping force is acting on the system for which $\gamma = 4\,Ns\,m^{-1}$ . Write down the differential equation of motion of the system . 1M

(OR)

4. Derive an equation for motion of a Longitudinal coupled oscillations and give a small note on normal coordinates & normal modes. 5M

5. (a) Write Conditions for Constructive & destructive Interferences 1M

(b) Explain how the refractive index of light source is determined by forming Newton's rings. 4M

(OR)

6. (a) Describe the principle, construction, theory and working of Michelson interferometer to find the wavelength and the difference in wavelength of a given light. 4M

(b) A combination of Plano convex lens and plane glass plate is illuminated by monochromatic source of light. The radius of 10th dark ring measured to be 0.28 cm. Find the wavelength of light used, if the radius of curvature of the lower face of the lens is 120 cm. 1M

**Rajiv Gandhi University of Knowledge Technologies- RK Valley- AP**

**Academic Year 2024-25: Semester II - Mid-I Examinations**

**CSE - BM3102 -MANAGERIAL ECONOMICS AND FINANCIAL ANALYSIS**

**R22 Batch & Regular Mode of Examinations**

**Year: ENGG-1** **Date of Examination: 31st Jan, 2025**

**Time: 60 Minutes** R220711 **Max Marks: 15**

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**The Question Paper consists of 1unit(s). Answer any one question each set.**
**Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions**

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1.What is Economics? How the managerial economics can be helpful in managerial economics? Explain.

**OR**

2. Define managerial economincs. Explain the nature of Managerial Economics.

3.Describe law of demand and its exceptions.

**OR**

4. What Is elasticity of Demand? Write about the types of Elasticity.

5. Explain supply function.

**OR**

6. Demonstrate the demand function

# Rajiv Gandhi University of Knowledge Technologies- RK Valley-AP

**Academic Year 2024 – 25: Semester II - Mid Semester Test-1**

**Computer Science Engineering (CSE) - 23MA1202 (Discrete Mathematics)(DM)**

**R22 Batch & Regular Mode of Examinations**

**Year: ENGG-1** | **Date of Examination: 01/02/2024**

**Time: 60 Minutes** R220711 | **Max Marks: 15**

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**3X5 Marks= 15**

**The Question Paper consists of 6 questions. Students need to answer 3 questions from each set**
**Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions**

1) Show that $[(p \vee q) \vee ((q \vee \sim r) \wedge (p \vee r))] \equiv (\sim(\sim p \wedge \sim q))$ without using truth tables.

**(OR)**

2) Find the truth table of $[p \rightarrow ((q \wedge \sim r) \vee s)] \wedge [\sim t \leftrightarrow (s \wedge r)]$

3) Find the PCNF and PDNF of $(p \wedge q) \vee (\sim p \wedge r) \vee (q \wedge r)$ without using truth tables.

**(OR)**

4) Test whether the following formula is a tautology (or) contradiction (or) contingency without using truth tables.
$[(p \vee q) \wedge \sim(\sim p \wedge (\sim q \vee \sim r))] \vee (\sim p \wedge \sim q) \vee (\sim p \wedge \sim r)$

5) Check the validity of the following argument.
If my program runs successfully then I will submit my project. I can appear in the examination only if I submit my project. Either my program runs successfully or the computer crashes. Therefore, if the computer does not crash then I can appear at the examination.

**(OR)**

6) Test whether the following premises are consistent (or) inconsistent.
$p \rightarrow (q \rightarrow r), s \rightarrow (q \wedge r), p \wedge s$